PORT VISITOR PASS MANAGEMENT SYSTEM

Business Analysis & System Requirements Specification

Client Solution & Functional Specification

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Prepared By	Tihan Liyanage
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VERSION HISTORY

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1.0	28 August 2026	Tihan Liyanage	Initial formal document generated from verified system architecture.

1. EXECUTIVE SUMMARY

The **Port Visitor Pass Management System (PORT PASS PRO)** is an enterprise-grade web application designed to automate, secure, and streamline the end-to-end lifecycle of visitor access within a port environment.

The system solves the operational problem of manual, fragmented visitor processing by providing a centralized platform where public visitors can request access, internal departments can approve visits, and security personnel can issue and monitor secure QR-coded passes. By implementing a strict state-machine workflow (Draft → Department Approval → Operations Review → Issued → Checked In/Out), the system ensures full accountability, eliminates unauthorized access, and maintains a comprehensive digital audit trail of all operational events. It is designed to be deployed securely on an office Local Area Network (LAN).

2. BUSINESS CONTEXT

Port environments require strict visitor access control to ensure safety, security, and compliance. The business context involves multiple operational layers: * **Public/External Access:** Visitors needing entry must submit formalized requests. * **Department Sponsorship:** Internal employees (hosts) or Department Managers must validate the business need for the visit. * **Operations & Security:** Central operations officers must verify identity (NIC/Passport) and background before authorizing access. * **Gate Control:** Physical entry points require fast, accurate verification (via QR scanning) to prevent bottlenecks while maintaining security. * **Auditability:** Management requires real-time visibility into who is currently on-site and historical records of all movements.

3. BUSINESS PROBLEM

Verified Existing Problems (Addressed by System): * Manual visitor processing and decentralized records. * Delayed inter-departmental approvals. * Difficulty tracking the real-time status of active passes (who is currently on-site). * Weak traceability of who approved a pass and when. * Lack of operational visibility into daily gate traffic.

TBC — Existing manual process requires client confirmation to baseline exact time/cost savings.

4. BUSINESS OBJECTIVES

- **BO-001:** Centralize all visitor requests into a single digital platform.
- **BO-002:** Standardize the multi-tier approval workflow (Department → Operations).
- **BO-003:** Improve gate monitoring via real-time QR code scanning and status validation.
- **BO-004:** Maintain a secure, immutable audit trail of all system actions.
- **BO-005:** Provide real-time reporting and analytics on visitor traffic and zone capacity.

5. STAKEHOLDER ANALYSIS

Stakeholder	Role	Responsibilities	System Interaction
Admin	System Administrator	Manages users, zones, capacities, and oversees system health.	Full access to all modules, logs, and settings.
Department Manager	Department Approver	Validates business need for visits addressed to their department.	Department Approvals Queue.
Operations Officer	Security/Ops Reviewer	Performs final validation, assigns visitor classifications (Normal/VIP), handles rejections.	Operations Approval Queue, Passes.
Pass Officer	Pass Issuer	Generates physical/digital passes with QR codes.	Pass Issuing, Pass Management, Group Passes.
Gate Officer	Access Controller	Scans passes at physical gates, logs check-ins and check-outs.	Gate Control Module.
Public Visitor	Requester	Submits requests for entry.	Public Visitor Request Portal.

6. USER ROLES AND RESPONSIBILITIES

1. Administrator (`admin`) * **Purpose:** Overall system governance. * **Permissions:** Unrestricted access across the platform, including Audit Logs and User Management.

2. Department Manager (`department_manager`) * **Purpose:** First-line approval. * **Permissions:** Can view and approve/reject requests targeted at their specific department.

3. Operations Officer (`operations_officer`) * **Purpose:** Security vetting. * **Permissions:** Reviews department-approved requests, assigns VIP/LMV status, provides final approval.

4. Pass Officer (`pass_officer`) * **Purpose:** Issuance. * **Permissions:** Generates passes, handles group passes, prints/distributes QR codes.

5. Security Gate Officer (`security_gate_officer`) * **Purpose:** Physical security enforcement. * **Permissions:** Can search passes, view visitor profiles, and trigger check-in/check-out events.

7. AS-IS PROCESS

TBC — Existing manual process requires client confirmation. Currently assumed to be paper-based or email-driven logbooks lacking real-time centralization.

8. TO-BE PROCESS

System-Enabled Workflow: 1. **Request:** Public visitor submits request via portal. (Status: `pending_department_approval`) 2. **Department Approval:** Manager approves. (Status: `pending_approval`) 3. **Operations Review:** Ops Officer verifies NIC and assigns zones/VIP status. (Status: `approved`) 4. **Pass Issuing:** Pass Officer generates QR pass. (Status: `issued`) 5. **Gate Check-In:** Visitor arrives, Gate Officer scans QR. (Status: `checked_in`) 6. **Gate Check-Out:** Visitor leaves, Gate Officer logs exit. (Status: `checked_out`)

9. BUSINESS PROCESS WORKFLOWS

9.1 Visitor Request

External users submit forms containing personal details, NIC/Passport, host employee, department, vehicle registration, and equipment details.

9.2 Department Approval

Managers view a filtered queue of requests targeting their department. They can approve (forward to Ops) or reject (terminate workflow).

9.3 Operations Approval

Officers review detailed identity profiles, assign classifications (e.g., VIP, LMV1/LMV2 routing), and provide final clearance.

9.4 Pass Issuing

Authorized officers generate secure passes containing hashed QR data and exact zone access arrays.

9.5 Visitor Check-In / 9.6 Check-Out

Gate officers use lookup or scanners to retrieve visitor profiles. The system strictly prevents check-in of unissued passes or check-out of visitors not checked in.

9.7 Audit Trail

Every state change, login, and data modification is asynchronously recorded in the `audit_logs` table.

10. FUNCTIONAL REQUIREMENTS

ID	Requirement	Priority	Role	Acceptance Criteria
FR-001	User Authentication	High	All Internal	Valid credentials yield a JWT token; invalid are rejected.
FR-002	Role-Based Access	High	All Internal	Users cannot access UI routes or APIs outside their assigned `UserRole`.
FR-003	Public Request Submission	High	Public	Unauthenticated users can submit a request and receive a tracking ID.
FR-004	Workflow State Enforcement	High	System	A pass cannot bypass the `pending_approval` state to become `issued` directly.
FR-005	QR Code Verification	High	Gate Officer	Scanning a valid QR code retrieves the current visitor profile and status.

FR-006	VIP & LMV Routing	Medium	Ops Officer	Ops can classify a visitor as VIP and specify LMV gate routing (LMV1/LMV2/Both).
FR-007	Zone Capacity Tracking	Medium	System	The dashboard must reflect current active visitors against predefined zone capacities.

11. MODULE-WISE REQUIREMENTS

Dashboard

- **Purpose:** High-level operational overview.
- **Outputs:** KPIs (Awaiting Issue, Currently Inside), Zone Capacity gauges, Daily Volume charts.

Visitor Requests (Public)

- **Purpose:** Data entry portal for visitors.
- **Validation:** Sri Lankan NIC format (9 digits + V/X or 12 digits), Phone formats.

Operations Approval

- **Purpose:** Final vetting stage.
- **Inputs:** Department-approved requests.
- **Outputs:** Approved or Rejected state, VIP flags.

Gate Control

- **Purpose:** Live gate operations.
- **Processing:** Checks current status. Prevents double-entry (checking in someone already checked in).

Audit Trail

- **Purpose:** Security compliance.
- **Processing:** Read-only ledger of actions (CREATE, UPDATE, DELETE, LOGIN, CHECK_IN).

12. DATA REQUIREMENTS

Entity	Purpose	Key Information
users	System accounts	Username, Password Hash, Role, Department_ID
visitor_requests	Tracks the request lifecycle	Requester details, NIC, Status, Workflow Timestamps
passes	Active/Historical passes	Pass No (QR), Zone array, Pass Type, Status
departments	Organizational units	Name, Manager Name
audit_logs	Security ledger	Action, Entity, User_ID, Timestamp, IP Address (Future)
zone_capacities	Occupancy limits	Zone ID, Max Capacity

13. BUSINESS RULES

- **BR-001:** A request must be approved by both a Department Manager and an Operations Officer before a pass can be issued.
- **BR-002:** A visitor cannot be checked out if they have not been checked in.
- **BR-003:** VIP visitors must be explicitly assigned LMV gate routing.
- **BR-004:** Pass Officers cannot approve requests; Operations Officers cannot issue passes (Segregation of Duties).

14. VALIDATION RULES

- **Identity:** NIC must match Sri Lankan format (e.g. 199012345678 or 901234567V).
- **Dates:** `requested_end_date` must be strictly greater than `requested_date`.
- **State Machine:** Check-in requires status `issued`. Check-out requires status `checked_in`.

15. ROLE AND ACCESS CONTROL MATRIX

Function	Admin	Dept Manager	Ops Officer	Pass Officer	Gate Officer
Dashboard	✓		✓	✓	✓ ✓
Dept Approvals	✓		✓		
Ops Approvals	✓			✓	
Issue Passes	✓				✓
Pass Management	✓				✓
Gate Control	✓				✓
Reports & Audit	✓				

16. NON-FUNCTIONAL REQUIREMENTS

- **NFR-001 Security:** Passwords must be hashed using bcrypt. APIs must be protected by JWT (JSON Web Tokens).
- **NFR-002 Performance:** The Gate Control lookup must return visitor details in under 1 second.
- **NFR-003 Usability:** The system must use a responsive CSS grid layout to support tablets used by gate officers.

17. MULTI-USER AND CONCURRENT ACCESS

The system is designed for concurrent multi-user environments. **Recommended LAN server architecture:** `text Dell Server PC (10.170.82.8) | +-- FastAPI (Port 8000) | +-- PostgreSQL (Port 5432) | +-- Vite Static Assets (Port 5173) | Office LAN (Wi-Fi/Ethernet) | +-----+-----+-----+ | | | HR PC Ops PC Gate Tablet` The architecture uses PostgreSQL with SQLAlchemy connection pooling, ensuring transaction safety when multiple officers process requests simultaneously.

18. SYSTEM ARCHITECTURE

- **Frontend:** React 18 Single Page Application, Vite, Ant Design & Material UI components, Zustand for state management.
- **Backend:** FastAPI (Python 3), Uvicorn ASGI server.
- **Database:** PostgreSQL 18.
- **Communication:** RESTful JSON APIs secured via JWT in Authorization headers.

19. DEPLOYMENT ARCHITECTURE

CURRENT / IMPLEMENTED: The system is deployed on a central office PC. * **Network:** LAN deployment accessible via `http://10.170.82.8:5173`. * **Firewall:** Windows Firewall rules explicitly allow inbound TCP on 5173 (Frontend) and 8000 (Backend). * **Startup:** Managed via batch scripts (`start_all.bat`).

20. SECURITY REQUIREMENTS

Currently Implemented: * JWT-based stateless authentication (token expires in 480 minutes). * Bcrypt password hashing. * CORS configured for network access. * SQLAlchemy ORM prevents SQL injection. * Strict RBAC enforcement on API endpoints.

RECOMMENDED Before Public Internet Production: * Implement HTTPS/TLS via Nginx reverse proxy. * Implement rate limiting to prevent brute force.

21. REPORTING AND ANALYTICS

- **Dashboard KPIs:** Live counts of pending requests, active passes, and visitors currently on-site.
- **Zone Analytics:** Visual pie charts detailing occupancy across physical zones.
- **Tabular Reports:** Paginated, filterable tables for historical pass data and audit logs.

22. AUDIT AND TRACEABILITY

The system maintains an immutable `audit_logs` table. Every significant state change (login, approval, check-in) generates a log containing the `user_id`, `action`, `entity_type`, and a UTC timestamp, ensuring full non-repudiation of officer actions.

23. USER EXPERIENCE

The UI provides a "Command Center" aesthetic with dark-mode optimized colors (Deep Navy, Cyan). Modals utilize clean CSS Grid layouts for equal-width data presentation. Empty states, skeleton loaders, and pulsing status dots provide continuous user feedback.

24. ASSUMPTIONS

- **ASM-001:** The server PC (Dell) remains powered on during operational hours.
- **ASM-002:** The office LAN provides stable connectivity to gate areas.
- **ASM-003: Client operational policies align with the system's strict sequential workflow. (TBC)**

25. CONSTRAINTS

- **Network Dependency:** Gate tablets must be connected to the office Wi-Fi network `10.170.82.8`.
- **Database:** Requires PostgreSQL; SQLite was deprecated for concurrency reasons.

26. RISKS

ID	Risk	Impact	Probability	Mitigation
R-01	Server PC goes offline	High	Medium	Ensure dedicated power/UPS for the server PC.
R-02	Wi-Fi dead zones at gates	High	Medium	Deploy Wi-Fi extenders at physical gate locations.

27. DEPENDENCIES

- PostgreSQL running on port 5432.
- Windows Firewall configured to allow ports 5173 and 8000.
- Modern web browsers (Chrome, Edge, Firefox) on client devices.

28. TESTING AND QUALITY ASSURANCE

Tested: * End-to-end SQLite to PostgreSQL data migration. * Concurrent multi-user session isolation. * UI Alignment across all detail modals (Ops, Dept, Gate). * Role-based route protection.

Not Yet Verified: * Physical QR code scanner hardware integration (currently supports manual ID entry as fallback).

29. PRODUCTION READINESS

Current Readiness: HIGH The system is functionally complete and successfully deployed on the LAN environment. There are no blocking software bugs. **Operational Prerequisites:** Staff training and distribution of the LAN URL and initial credentials.

30. ACCEPTANCE CRITERIA

- **AC-001:** Users can only log in with valid credentials and are routed to their role-specific dashboard.
- **AC-002:** A department manager can only see requests targeting their specific department ID.
- **AC-003:** The system correctly formats Sri Lankan NICs and displays validation errors for incorrect lengths.
- **AC-004:** The Gate Control dashboard accurately toggles a pass status between `checked_in` and `checked_out`.

31. REQUIREMENTS TRACEABILITY MATRIX

BO-001 (Centralize) → FR-003 (Public Request) → VisitorRequest Module → AC-003 (NIC Validation)
BO-004 (Audit) → FR-007 (Action Logging) → AuditLogs Module → AC-005 (Log generation on approval)

32. FUTURE ENHANCEMENTS

Future / Proposed: * Email/SMS notifications to visitors upon pass approval. * Active Directory / LDAP integration for internal employee logins. * Automated ID/Passport OCR scanning at the gate.

33. BUSINESS BENEFITS

- **Operational Efficiency:** Eliminates physical paper trails and inter-office phone calls for approvals.
- **Security:** Cryptographically secure QR codes prevent forgery.
- **Accountability:** Every approval and gate scan is permanently linked to a specific officer.
- **Real-time Visibility:** Management instantly knows exact on-site visitor counts during emergencies.

34. IMPLEMENTATION ROADMAP

1. **Phase 1:** PostgreSQL Migration & LAN Deployment Setup (Completed)
2. **Phase 2:** Enterprise UI Redesign (Completed)
3. **Phase 3:** Staff Training & Credentials Handoff
4. **Phase 4:** Parallel Run (Digital + Manual)
5. **Phase 5:** Full Production Launch

35. TRAINING REQUIREMENTS

- **Gate Officers:** Require training on scanning passes and interpreting the "VIP LMV Routing" banners.
- **Department Managers:** Require instruction on accessing their queue via browser and adding approval notes.

36. SUPPORT AND MAINTENANCE

Recommended daily backups of the `port_pass_db` PostgreSQL database using `pg_dump`. Server OS requires standard security patching.

37. GLOSSARY

- **RBAC:** Role-Based Access Control.
- **JWT:** JSON Web Token used for secure sessions.
- **LMV:** Light Motor Vehicle (Gate routing parameter).
- **QR:** Quick Response code containing hashed validation data.

38. APPENDICES

- **Appendix A:** Database Schema (Refer to SQLAlchemy models)
- **Appendix B:** Office LAN Setup Guide (Refer to `deploy_lan.bat`)